AIAA Guidance, Navigation and Control Conference and Exhibit
18 - 21 August 2008, Honolulu, Hawaii

AIAA 2008-6454

Estimation of Spacecraft Inertia Parameters

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The rigid body dynamics of a spacecraft are dependent on the spacecraft inertia. The inertia matrix is determined prior to flight during the spacecraft development. However, the inertia properties can change after launch due to the shifting of hardware caused by launch or environmental effects, fuel usage, or failure of a component to deploy into the expected configuration. This paper presents a method for adaptively estimating all the spacecraft inertia components, with no a priori information, as part of a passivity-based angular velocity tracking control algorithm. A persistency of excitation condition guarantees exponential convergence of the inertia estimates to the true values.

I. Introduction

The problem of inertia estimation has been treated by numerous authors, through a variety of control and estimation methods. Tanygin and Williams[1] develop a least squares approach to estimate the inertia parameters based on the rotational kinetic energy. They recognize the need for a persistently exciting signal and connect it loosely to geometrical considerations. Psiaki[2] solves a complex, total least squares problem to estimate the inertia matrix, as well as actuator calibration components. The method relies on attitude and rate measurements and is applied to real spacecraft data, without an actual truth model. The computationally intensive algorithm performs well and reduces the spacecraft torque modeling error.

Chaturvedi, et al[3] present an adaptive control algorithm in which asymptotic identification of the inertia parameters is achieved with sinusoidal angular velocity profiles. They use a feedback linearization type approach to demonstrate the convergence properties of their method.

In this work we apply a nonlinear passivity-based adaptive control strategy to angular velocity tracking for a spacecraft with unknown inertia parameters. As in [3], the passivity-based control algorithm tracks a desired, continuously differentiable, angular velocity. The passivity-based approach, however, reduces the order of the dependence on the measured angular velocity from quadratic to linear. The control scheme presented here is similar to that of Luquette,[4] with only the angular velocity tracking under consideration. With perfect knowledge of the spacecraft inertia, the control scheme is exponentially stable. Expanding the control to adapt to unknown inertia parameters, as in [4], results in the tracking error converging asymptotically to zero, without a guarantee of identifying the inertia parameters. However, if the angular velocity meets a persistency of excitation condition, both the tracking error and the inertia estimation error converge exponentially to zero. The persistency of excitation condition results in a 6x6 positive definite matrix. The minimum eigenvalue of the persistency of excitation matrix defines the exponential convergence

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rate. We prove the persistency of excitation condition and present simulation results that demonstrate the convergence properties based on the minimum eigenvalue of the persistency of excitation matrix.

II. Adaptive Control Algorithm

The angular velocity dynamics of a rigid spacecraft are

$$\boldsymbol{H}\dot{\boldsymbol{\omega}} - [\boldsymbol{H}\boldsymbol{\omega}\times]\boldsymbol{\omega} = \boldsymbol{\tau} \tag{1}$$

where $\boldsymbol{H}$ is the 3x3 symmetric moment of inertia matrix, resolved in body coordinates, $\boldsymbol{\omega}$ is the spacecraft angular velocity vector, and $\boldsymbol{\tau}$ is the external torque vector, all resolved in spacecraft body coordinates. $[\cdot\times]$ is the skew symmetric cross product matrix. The objective is to force $\boldsymbol{\omega}(t)$ to track a desired angular velocity profile, $\boldsymbol{\omega}_d(t)$, with $\boldsymbol{\omega}(t)$ measured perfectly. Define the tracking error $\tilde{\boldsymbol{\omega}}(t) = \boldsymbol{\omega}(t) - \boldsymbol{\omega}_d(t)$, then

$$\begin{aligned} \boldsymbol{H}\dot{\tilde{\boldsymbol{\omega}}}(t) &= \boldsymbol{H}\dot{\boldsymbol{\omega}}(t) - \boldsymbol{H}\dot{\boldsymbol{\omega}}_d(t) \\ &= \boldsymbol{\tau}(t) + [\boldsymbol{H}\boldsymbol{\omega}(t)\times]\boldsymbol{\omega}(t) - \boldsymbol{H}\dot{\boldsymbol{\omega}}_d(t) \end{aligned}$$

If the torque $\boldsymbol{\tau}(t)$ is chosen as

$$\boldsymbol{\tau}(t) = \boldsymbol{H}\dot{\boldsymbol{\omega}}_d(t) - [\boldsymbol{H}\boldsymbol{\omega}(t)\times]\boldsymbol{\omega}_d(t) - \boldsymbol{\Gamma}\tilde{\boldsymbol{\omega}}(t) \tag{2}$$

with $\boldsymbol{\Gamma}$ any positive definite, symmetric matrix, then the tracking error $\tilde{\boldsymbol{\omega}}(t)$ will converge exponentially to zero. To show this, substitute 2 into 1 to obtain the closed-loop dynamics

$$\boldsymbol{H}\dot{\tilde{\boldsymbol{\omega}}}(t) - [\boldsymbol{H}\boldsymbol{\omega}(t)\times]\tilde{\boldsymbol{\omega}}(t) + \boldsymbol{\Gamma}\tilde{\boldsymbol{\omega}}(t) = 0$$

The Lyapunov function $V = \frac{1}{2}\tilde{\boldsymbol{\omega}}^T\boldsymbol{H}\tilde{\boldsymbol{\omega}}$ is easily shown to have time derivative $\dot{V} = -\tilde{\boldsymbol{\omega}}^T\boldsymbol{\Gamma}\tilde{\boldsymbol{\omega}} \leq 0$ thereby proving the exponential convergence property.[5]

In practice, $\boldsymbol{H}$ is not known exactly and must be estimated on-orbit. Let the vector $\boldsymbol{h}$ consist of the 6 unique elements of the symmetric inertia matrix

$$\boldsymbol{h} = \begin{bmatrix} H_{11} \\ H_{22} \\ H_{33} \\ H_{12} \\ H_{13} \\ H_{23} \end{bmatrix}$$

The input torque above can be written as

$$\boldsymbol{\tau}(t) = \boldsymbol{Y}(\boldsymbol{\omega}(t), \boldsymbol{\omega}_d(t), \dot{\boldsymbol{\omega}}_d(t))\boldsymbol{h} - \boldsymbol{\Gamma}\tilde{\boldsymbol{\omega}}(t)$$

where $\boldsymbol{Y}$ is the 3×6 matrix

$$\boldsymbol{Y}(\boldsymbol{\omega}(t), \boldsymbol{\omega}_d(t), \dot{\boldsymbol{\omega}}_d(t)) = \dot{\boldsymbol{\Omega}}_d(t) + [\boldsymbol{\omega}_d(t)\times]\boldsymbol{\Omega}(t) \tag{3}$$

where

$$\dot{\boldsymbol{\Omega}}_d = \begin{bmatrix} \dot{\omega}_{d,1} & 0 & 0 & \dot{\omega}_{d,2} & \dot{\omega}_{d,3} & 0 \\ 0 & \dot{\omega}_{d,2} & 0 & \dot{\omega}_{d,1} & 0 & \dot{\omega}_{d,3} \\ 0 & 0 & \dot{\omega}_{d,3} & 0 & \dot{\omega}_{d,1} & \dot{\omega}_{d,2} \end{bmatrix}$$

and, similarly

$$\boldsymbol{\Omega} = \begin{bmatrix} \omega_1 & 0 & 0 & \omega_2 & \omega_3 & 0 \\ 0 & \omega_2 & 0 & \omega_1 & 0 & \omega_3 \\ 0 & 0 & \omega_3 & 0 & \omega_1 & \omega_2 \end{bmatrix}$$

$(\dot{\omega}_{d,1}, \dot{\omega}_{d,2}, \dot{\omega}_{d,3})$ and $(\omega_1, \omega_2, \omega_3)$ are the components of $\dot{\boldsymbol{\omega}}_d(t)$ and $\boldsymbol{\omega}(t)$, respectively. Since $\boldsymbol{h}$ is not known, we use estimates $\hat{\boldsymbol{h}}$ in place of $\boldsymbol{h}$, thus implementing

$$\boldsymbol{\tau}(t) = \boldsymbol{Y}(\boldsymbol{\omega}(t), \boldsymbol{\omega}_d(t), \dot{\boldsymbol{\omega}}_d(t))\hat{\boldsymbol{h}} - \boldsymbol{\Gamma}\tilde{\boldsymbol{\omega}}(t)$$

producing the passive closed-loop dynamics

$$\boldsymbol{H}\dot{\tilde{\boldsymbol{\omega}}}(t) - [\boldsymbol{H}\boldsymbol{\omega}(t)\times]\tilde{\boldsymbol{\omega}}(t) + \boldsymbol{\Gamma}\tilde{\boldsymbol{\omega}}(t) = \boldsymbol{Y}(\boldsymbol{\omega}(t), \boldsymbol{\omega}_d(t), \dot{\boldsymbol{\omega}}_d(t))\tilde{\boldsymbol{h}} \tag{4}$$

where $\tilde{\boldsymbol{h}} = \hat{\boldsymbol{h}} - \boldsymbol{h}$.

To derive a stable online estimation strategy for the unknown inertias, consider the Lyapunov function

$$V_1 = \frac{1}{2}\tilde{\boldsymbol{\omega}}^T \boldsymbol{H}\tilde{\boldsymbol{\omega}} + \frac{1}{2\gamma}\tilde{\boldsymbol{h}}^T\tilde{\boldsymbol{h}}$$

where γ is any positive constant. The time derivative of V_1 is along the solution of the closed-loop dynamics given in equation 4

$$\dot{V}_1 = -\tilde{\boldsymbol{\omega}}^T\boldsymbol{\Gamma}\tilde{\boldsymbol{\omega}} + (\tilde{\boldsymbol{\omega}}^T\boldsymbol{Y} + \frac{1}{\gamma}\dot{\tilde{\boldsymbol{h}}}^T)\tilde{\boldsymbol{h}}$$

Since $\boldsymbol{h}$ is constant, $\dot{\tilde{\boldsymbol{h}}} = \dot{\hat{\boldsymbol{h}}}$ and the choice

$$\dot{\hat{\boldsymbol{h}}} = -\gamma\boldsymbol{Y}^T\tilde{\boldsymbol{\omega}} \tag{5}$$

renders $\dot{V}_1$ negative semi-definite

$$\dot{V}_1 = -\tilde{\boldsymbol{\omega}}^T\boldsymbol{\Gamma}\tilde{\boldsymbol{\omega}} \leq 0$$

The closed-loop dynamics are proven stable with this estimation strategy. Application of Barbalat's lemma additionally shows that $\tilde{\boldsymbol{\omega}}(t)$ converges to zero asymptotically.[6]

This strategy solves the angular velocity tracking problem in the presence of inertia uncertainty. Nothing in the above argument, however, ensures that the inertia components are completely identified. That is, it is not necessarily true that $\tilde{\boldsymbol{h}}(t) \to \boldsymbol{0}$, unless additional conditions are satisfied.

We have established that the angular velocity tracking error, $\tilde{\boldsymbol{\omega}}$, converges to zero. Therefore, the rate of the inertia estimate, given in (5), also converges to zero, indicating that $\tilde{\boldsymbol{h}}$ converges at least to a constant. From the closed loop dynamics given in (4)

$$\boldsymbol{Y}(\boldsymbol{\omega}(t), \boldsymbol{\omega}_d(t), \dot{\boldsymbol{\omega}}_d(t))\tilde{\boldsymbol{h}} \to \boldsymbol{0} \tag{6}$$

Accurate identification of the inertia requires the *persistency of excitation* (PE) condition: there exists a $T_1 \geq 0$ and a $T_2 > 0$ such that the 6×6 matrix

$$\boldsymbol{P}(t) = \int_t^{t+T_2} \boldsymbol{Y}(\boldsymbol{\omega}(\tau), \boldsymbol{\omega}_d(\tau), \dot{\boldsymbol{\omega}}_d(\tau))^T \boldsymbol{Y}(\boldsymbol{\omega}(\tau), \boldsymbol{\omega}_d(\tau), \dot{\boldsymbol{\omega}}_d(\tau))d\tau$$

is positive definite for any $t \geq T_1$. Multiplying $\boldsymbol{P}(t)$ by a constant $\tilde{\boldsymbol{h}}$, with (6)

$$\left[\int_t^{t+T_2} \boldsymbol{Y}(\boldsymbol{\omega}(\tau), \boldsymbol{\omega}_d(\tau), \dot{\boldsymbol{\omega}}_d(\tau))^T \boldsymbol{Y}(\boldsymbol{\omega}(\tau), \boldsymbol{\omega}_d(\tau), \dot{\boldsymbol{\omega}}_d(\tau))d\tau\right]\tilde{\boldsymbol{h}} \to 0$$

If $\boldsymbol{P}(t)$ is positive definite, the above implies that $\tilde{\boldsymbol{h}} \to 0$.[7]

Evaluation of $\boldsymbol{P}(t)$ requires knowledge of the closed-loop solution for $\boldsymbol{\omega}(t)$ which will depend on the (initially) unknown inertias. From the definition of $\tilde{\boldsymbol{\omega}}$, define $\tilde{\boldsymbol{\Omega}}$ as

$$\tilde{\boldsymbol{\Omega}}(t) \equiv \boldsymbol{\Omega}(t) - \boldsymbol{\Omega}_d(t)$$

Substituting for $\boldsymbol{\Omega}(t)$ in equation 3 results in

$$\boldsymbol{Y}(\boldsymbol{\omega}(t), \boldsymbol{\omega}_d(t), \dot{\boldsymbol{\omega}}_d(t)) = \dot{\boldsymbol{\Omega}}_d(t) + [\boldsymbol{\omega}_d(t)\times]\left(\boldsymbol{\Omega}_d(t) + \tilde{\boldsymbol{\Omega}}(t)\right)$$

Since the above proof ensures that $\boldsymbol{\omega}(t) \to \boldsymbol{\omega}_d(t)$ asymptotically, independent of the PE condition, for sufficiently large T_1, $\boldsymbol{\Omega}(t) \to \boldsymbol{\Omega}_d(t)$, implies that

$$\boldsymbol{Y}(\boldsymbol{\omega}(t), \boldsymbol{\omega}_d(t), \dot{\boldsymbol{\omega}}_d(t)) \to \boldsymbol{Y}_d(\boldsymbol{\omega}_d(t), \dot{\boldsymbol{\omega}}_d(t))$$

where

$$\boldsymbol{Y}_d(\boldsymbol{\omega}_d(t), \dot{\boldsymbol{\omega}}_d(t)) = \dot{\boldsymbol{\Omega}}_d(t) + [\boldsymbol{\omega}_d(t)\times]\boldsymbol{\Omega}_d(t)$$

Therefore, the PE condition is equivalent to the positive definiteness of

$$\boldsymbol{P}_d(t) = \int_t^{t+T_2} \boldsymbol{Y}_d(\boldsymbol{\omega}_d(\tau), \dot{\boldsymbol{\omega}}_d(\tau))^T \boldsymbol{Y}_d(\boldsymbol{\omega}_d(\tau), \dot{\boldsymbol{\omega}}_d(\tau)) d\tau \tag{7}$$

which can be evaluated solely in terms of the *desired* angular velocity $\boldsymbol{\omega}_d(t)$ the vehicle is commanded to follow. Thus, by carefully constructing an angular velocity profile $\boldsymbol{\omega}_d(t)$ to ensure positive definiteness of $\boldsymbol{P}_d(t)$, the estimation and control algorithm above will accurately identify the unknown spacecraft inertia components.

We have now established that $\tilde{\boldsymbol{h}} \to \boldsymbol{0}$, at least asymptotically. Because all the signals are bounded, the above system can also be represented mathematically as a linear time varying system (See [5], pp. 626-628).

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}(t)\boldsymbol{x}(t)$$

where $\boldsymbol{x}^T = \begin{bmatrix} \tilde{\boldsymbol{\omega}}^T & \tilde{\boldsymbol{h}}^T \end{bmatrix}$ and $\dot{V}_1$ is rewritten as $\dot{V}_1 = -\boldsymbol{x}^T \boldsymbol{C}^T \boldsymbol{C}\boldsymbol{x} \le 0$, with $\boldsymbol{C} \equiv \begin{bmatrix} \sqrt{\boldsymbol{\Gamma}} & \boldsymbol{0} \end{bmatrix}$. Theorem 4.5 and the discussion on pp. 626-628 in [5] show that the equilibrium point $\boldsymbol{x} = \boldsymbol{0}$ is exponentially stable if the pair $(\boldsymbol{A}(t), \boldsymbol{C})$ is uniformly completely observable. The PE condition ensures uniform complete observability of the pair $(\boldsymbol{A}(t) - \boldsymbol{K}(t)\boldsymbol{C}, \boldsymbol{C})$, with $\boldsymbol{K}(t)$ chosen such that it is piecewise continuous and bounded. Since observability properties are unchanged under output feedback, this ensures uniform complete observability of the pair $(\boldsymbol{A}(t), \boldsymbol{C})$. Moreover, the convergence of both the inertia error and the tracking error will now be exponential, as opposed to simply asymptotic ([8], pp. 326-327).

Persistency of Excitation

An example angular velocity that satisfies the PE condition is

$$\boldsymbol{\omega}_d^T = a \begin{bmatrix} \sin \nu t & \cos \nu t & \sin 2\nu t \end{bmatrix}$$

Substituting $\boldsymbol{\omega}_d$ into equation 7, and integrating with $T_2 = \frac{2\pi}{\nu}$ results in the positive definite $\boldsymbol{P}_d$

$$\boldsymbol{P}_d = \frac{2\pi a^2}{\nu} \begin{bmatrix} \frac{\nu^2}{2} + \frac{3a^2}{8} & -\frac{a^2}{8} & -\frac{a^2}{4} & 0 & 0 & 0 \\ -\frac{a^2}{8} & \frac{\nu^2}{2} + \frac{3a^2}{8} & -\frac{a^2}{8} & 0 & 0 & 0 \\ -\frac{a^2}{4} & -\frac{a^2}{8} & 2\nu^2 + \frac{a^2}{2} & -\frac{a\nu^3}{2} & 0 & 0 \\ 0 & 0 & -\frac{a\nu^3}{2} & \nu^2 + a^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5(\frac{\nu^2}{2} + \frac{a^2}{8}) & \frac{a\nu^3}{2} \\ 0 & 0 & 0 & 0 & \frac{a\nu^3}{2} & 5(\frac{\nu^2}{2} + \frac{a^2}{8}) \end{bmatrix}$$

If $a = 1$ rad/sec and $\nu = 1$ rad/sec, $\boldsymbol{P}_d$ is

$$\boldsymbol{P}_d = \begin{bmatrix} 5.5 & -0.79 & -1.57 & 0 & 0 & 0 \\ -0.79 & 5.5 & -1.57 & 0 & 0 & 0 \\ -1.57 & -1.57 & 15.7 & -9.42 & 0 & 0 \\ 0 & 0 & -9.42 & 12.57 & 0 & 0 \\ 0 & 0 & 0 & 0 & 19.64 & 9.42 \\ 0 & 0 & 0 & 0 & 9.42 & 19.64 \end{bmatrix}$$

The minimum eigenvalue is 3.14.

The exponential convergence rate of $\tilde{\boldsymbol{h}}$ is proportional to the lower bound of the observability Gramian of the pair $(\boldsymbol{A}(t), \boldsymbol{C})$.[8] However, the lower bound of the observability Gramian for $(\boldsymbol{A}(t), \boldsymbol{C})$ is proportional to the lower bound of the observability Gramian of the equivalent system $(\boldsymbol{A}(t) - \boldsymbol{K}(t)\boldsymbol{C}, \boldsymbol{C})$.[9,10] In [10], Sastry and Bodson develop a relationship between the exponential convergence rate defined by the UCO properties of $(\boldsymbol{A}(t) - \boldsymbol{K}(t)\boldsymbol{C}, \boldsymbol{C})$ and the bounds on the PE matrix. We demonstrate this relationship by varying the eigenvalues of $\boldsymbol{P}_d$. As a and ν increase and decrease, the minimum eigenvalue simultaneously increases and decreases. If $a = 0.5$ and $\nu = 0.5$ the minimum eigenvalue is reduced to 0.39, resulting in a longer convergence time. The amplitude and frequency of the desired angular velocity can be set based on mission or vehicle constraints, such as available torque authority.

III. Results

The first scenario tested consists of an arbitrary inertia matrix for a moderately sized spacecraft.

$$\boldsymbol{H} = \begin{bmatrix} 200 & 10 & 50 \\ 10 & 150 & -25 \\ 50 & -25 & 100 \end{bmatrix} \text{kg-m}^2$$

The desired angular velocity is again given by

$$\boldsymbol{\omega}_d^T = a \begin{bmatrix} \sin \nu t & \cos \nu t & \sin 2\nu t \end{bmatrix}$$

Table 1 lists the magnitude and frequency for three test cases. Also included in the table is the minimum eigenvalue of $\boldsymbol{P}_d$ for each magnitude and frequency. The a priori inertia estimates are all initialized to

Table 1. Magnitude, Frequency and Minimum Eigenvalues

Case	Magnitude (rad/sec)	Frequency (rad/sec)	λ_{min}
1	1	1	3.14
2	0.3	0.3	0.08
3	0.2	0.2	0.02

zero, $\hat{\boldsymbol{h}}(0) = \mathbf{0}$. Figure 1 shows the convergence of the six inertia components for each set of magnitudes and frequency. The gains are $\boldsymbol{\Gamma} = 50\mathrm{I}_3$kg-m^2/sec and $\gamma = 100$ kg-m^2-sec. The convergence time decreases dramatically as the minimum eigenvalue of $\boldsymbol{P}_d$ increases.

The above scenarios are repeated with a much larger inertia matrix, that of the Hubble Space Telescope[11]

$$\boldsymbol{H} = \begin{bmatrix} 36046 & -706 & 1491 \\ -706 & 86868 & 449 \\ 1491 & 449 & 93848 \end{bmatrix} \text{kg-m}^2$$

The gains are increased proportionately to the size of the inertia matrix to $\boldsymbol{\Gamma} = 2 \times 10^4 \mathrm{I}_3$ kg-m^2/sec and $\gamma = 4 \times 10^4$ kg-m^2-sec. The results are given in figure 2.

IV. Conclusion

A nonlinear passivity adaptive control scheme is utilized to fully estimate the six inertia parameters of a rigid spacecraft. The estimation is achieved by selecting a desired angular velocity profile which meets a persistency of excitation condition. The persistency of excitation condition results in a positive definite matrix. The minimum eigenvalue of this matrix defines the exponential convergence rate of the estimated inertia parameters to the true inertia parameters. Two sample simulations are provided to demonstrate the estimation for both a moderate sized spacecraft and a very large spacecraft. Future work will address how the algorithm behaves in the presence of uncertainties and noise. It is anticipated that the passivity approach will offer robustness as compared to other adaptive control approaches.

V. Acknowledgment

This research was funded by a Naval Academy Research Center grant. J.K. Thienel also thanks Mr. Rick Harman of the NASA Goddard Space Flight Center for identifying and requesting the research.

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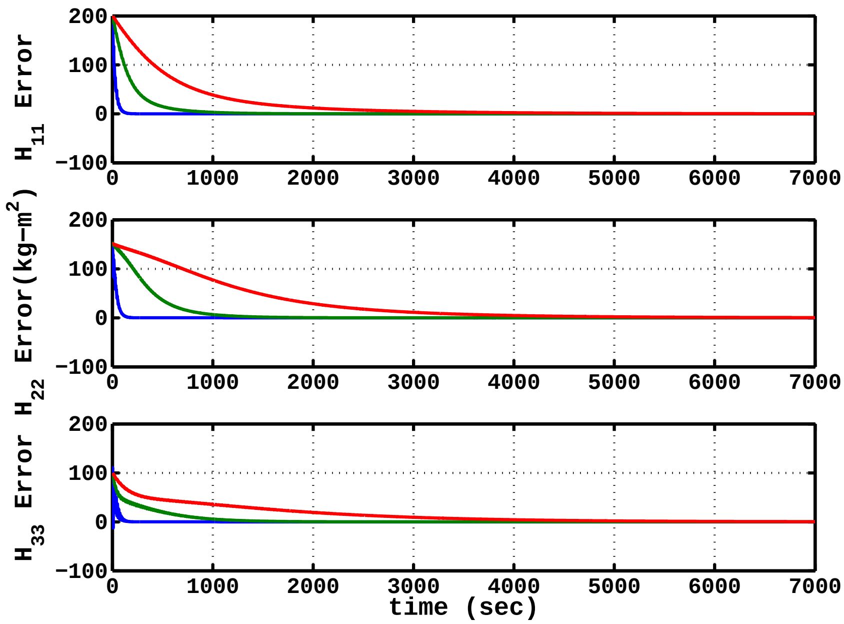
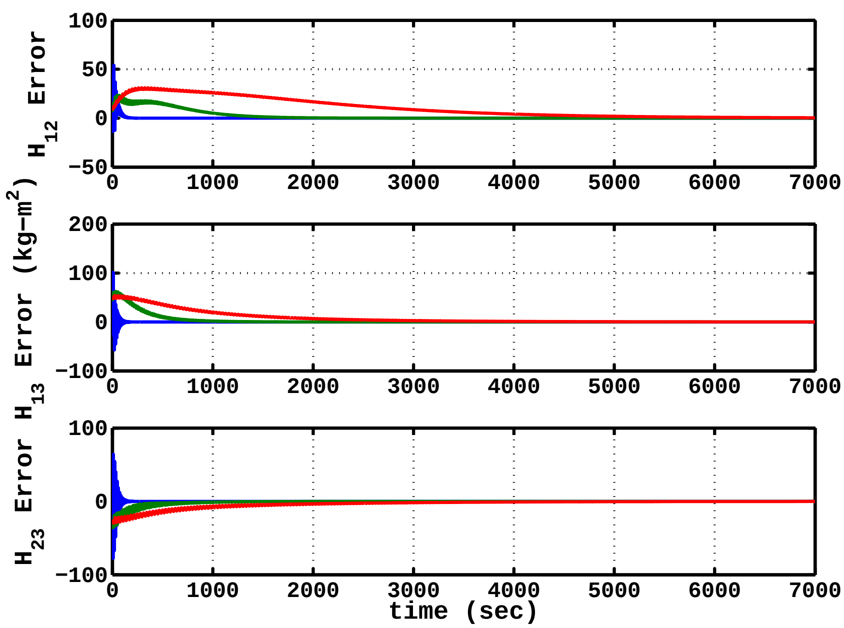
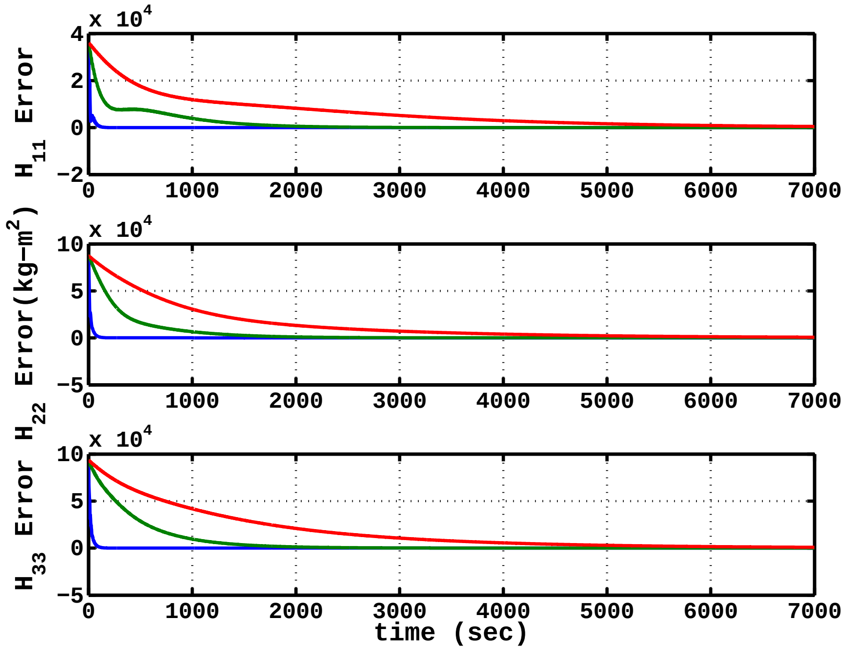


(a) Error in Diagonal Elements, Case 1 (blue), Case 2 (green), Case 3(red)

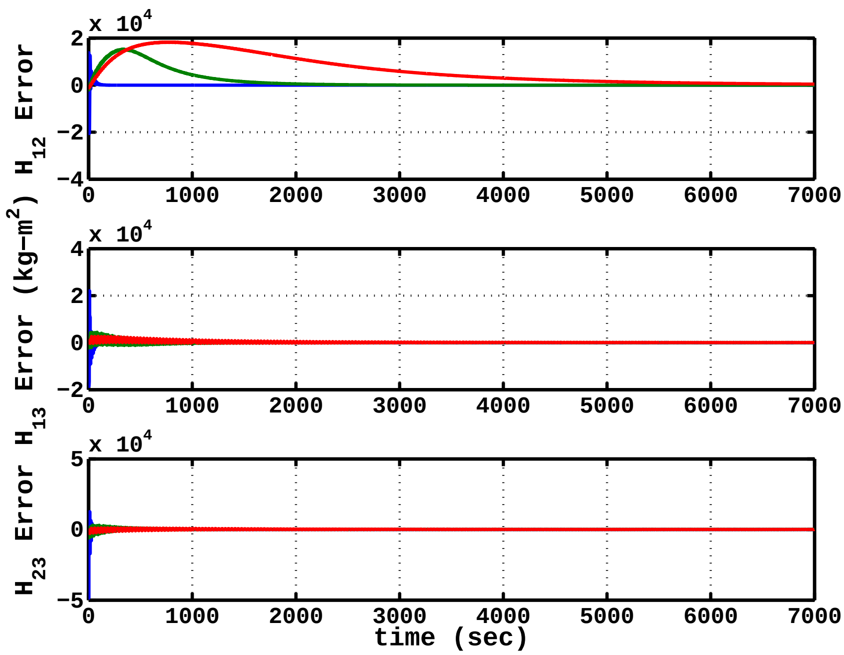


(b) Error in Diagonal Elements, Case 1 (blue), Case 2 (green), Case 3(red)

Figure 1. Inertia Estimation Errors



(a) Error in Diagonal Elements, Case 1 (blue), Case 2 (green), Case 3(red)



(b) Error in Diagonal Elements, Case 1 (blue), Case 2 (green), Case 3(red)

Figure 2. Inertia Estimation Errors, Large Inertia Matrix